

SOLUTIONS

2.1-Practice:

1. Find the horizontal and vertical asymptotes for the following functions.

a. $f(x) = \frac{2x^2(x^2-1)}{(x+2)^2(x^2-4)}$

b. $f(x) = \frac{2(x-3)(x+2)(x+5)}{(x-1)(x+3)(x+5)}$

V.A: $(x+2)^2=0$ $x^2-4=0$
 $x = -2$ $x = \pm 2$

V.A: $x-1=0$ $x+3=0$
 $x = 1$ $x = -3$

$\therefore$ V.A are $x=2$ and $x=-2$

$\therefore$ V.A are $x=1, x=-3$

H.A: $f(x) = \frac{2x^4 + \dots}{x^4 + \dots}$

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$\therefore$ H.A at $y=2$

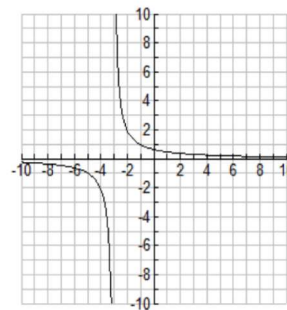
$\therefore$ H.A at $y=2$

2. Determine the equation of the following graph:

V.A: $x=-3$ H.A: $y=0$

$f(x) = \frac{k}{x+b}$

Sub. $(-1, 1)$: $1 = \frac{k}{-1+3}$ $\therefore f(x) = \frac{2}{x+3}$
 $1 = \frac{k}{2}$
 $2 = k$



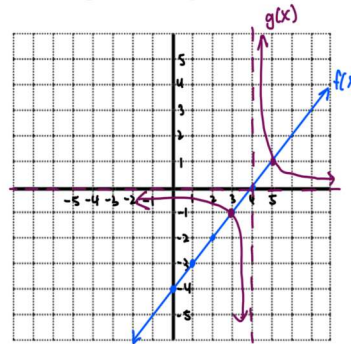
3. Graph $f(x) = -(4-x)$ and $g(x) = \frac{-1}{4-x}$ on the grid provided and for the $g(x)$ identify:

a) the domain $\{x \in \mathbb{R} \mid x \neq 4\}$

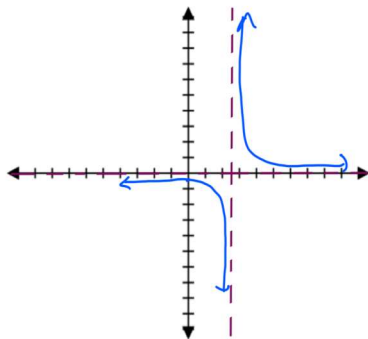
b) the range $\{y \in \mathbb{R} \mid y \neq 0\}$

c) the equation of the V.A. $x=4$

d) the equation of the H.A. $y=0$



4. Determine the following information and sketch graph.



Equation	$y = \frac{1}{2x-5}$		
Domain	$\{x \in \mathbb{R} \mid x \neq \frac{5}{2}\}$		
Range	$\{y \in \mathbb{R} \mid y \neq 0\}$		
x-int	None	y-int	$-\frac{1}{5}$
H. Asymptote	$y=0$	V. Asymptote	$x = \frac{5}{2}$

As $x \rightarrow$	$f(x) \rightarrow$
$\frac{5}{2}^+$	∞
$\frac{5}{2}^-$	$-\infty$
$+\infty$	0 (above)
$-\infty$	0 (below)